
535510 RL Team Project Midterm Report: Reinforcement Learning for Scenic Photo Parameter Adjustment with TOPIQ Reward

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1 Project Overview

1.1 Profile

Track: Application

Abstract and project goal: This project focuses on enhancing the aesthetic quality of scenic photos by adjusting parameters like contrast, brightness, saturation, and gamma. Using Reinforcement Learning (RL), specifically Proximal Policy Optimization (PPO), the agent learns to optimize these parameters based on human aesthetic preferences, transforming the photo to make it more visually appealing. The enhancement process is guided by feedback from TOPIQ, a no-reference image quality metric, which evaluates the improvement in visual quality. During training, the agent utilizes histogram-based features and a CNN-based network to extract low-level image features, enabling the RL model to make more precise adjustments.

TL;DR ("Too Long; Didn't Read"): This project uses RL to adjust the parameters of scenic photos, optimizing them to better match human aesthetic preferences, with feedback provided by TOPIQ as the reward function.

1.2 Motivation

1. Why is the problem interesting?

Capturing scenic photos often doesn't result in the ideal image right away, requiring post-processing adjustments. When tweaking parameters like color, contrast, brightness, and saturation, they often interact with one another — for instance, adjusting brightness can lead to unnatural color contrast. This makes it a suitable problem for solving with Reinforcement Learning, where parameters can be adjusted incrementally and optimized based on the reward. We found that the TOPIQ [Chen et al., 2024] offers an image quality evaluation, making it an ideal reward function for RL applications.

2. Challenges

- **Challenge 1: The Limitations of Using a Single Reward Metric**
Adjusting multiple aspects of a photo can lead to conflicts between parameters, causing non-converging rewards when using only TOPIQ. If issues arise, we may consider incorporating additional metrics like SSIM and PSNR to guide convergence.
- **Challenge 2: Adjusting Multiple Parameters Simultaneously**
When adjusting parameters like color, brightness, and saturation together, they interact in complex ways, making it harder for the RL agent to find the optimal balance. This could lead to unwanted artifacts. A similar approach is used in [Wu et al., 2024], where parameters are

adjusted sequentially, and we can learn from this method to handle simultaneous adjustments more effectively.

- **Challenge 3: Limited Photometric Sensitivity of Frozen Encoder.**

ResNet-50 was originally trained for ImageNet classification rather than image quality assessment, so its frozen representations may lack sensitivity to low-level photometric properties such as brightness and contrast that are critical for enhancement decisions. Similarly, intensity and gradient histograms capture only global statistics and remain insensitive to local spatial variations, potentially limiting the agent’s ability to select appropriate actions for spatially non-uniform images.

3. Justify why the problem remains open or unsolved?

- **Limitations of Real-Time RL for Single Parameter Adjustment**

Many RL-based approaches focus on adjusting specific camera parameters. For example, the method in [Lee et al., 2024] focuses on controlling exposure, while [Lee et al., 2026] targets automatic white balance correction. These methods emphasize the real-time nature of the task, often training the RL model to adjust a single parameter, such as exposure or white balance. However, this approach may not be sufficient for creating images that align with the user’s aesthetic preferences, as adjusting only one parameter might not achieve the desired visual appeal.

- **Need for Dataset Creation to Evaluate Optimal Parameter Adjustment**

In DRL-ISP [Shin et al., 2022], the model is trained using correctly paired ground truth (GT) images and their unadjusted counterparts, where GT images serve as the target for the RL agent to match through parameter adjustments. However, this necessitates the creation of a large labeled dataset consisting of GT $\leftrightarrow$ original photo pairs, and the reward can become unstable if image styles vary across the dataset, making it difficult to define universally “good” adjustments.

4. The state-of-the-art method

DRL-ISP [Shin et al., 2022] trains an RL model to optimize image processing tools for tasks like deblurring, denoising, and exposure correction. The model uses histograms and CNNs to extract low-level features, and PPO to adjust parameters based on task performance. We find that our approach is similar to DRL-ISP because both involve using reinforcement learning to adjust image parameters. Specifically, we can draw from their use of CNNs for feature extraction and PPO for reinforcement learning to optimize our image adjustments.

However, unlike DRL-ISP, which focuses on functional restoration and requires large labeled datasets, our method aims to enhance the aesthetic appeal of an image by adjusting parameters such as contrast, saturation, and brightness according to human aesthetic preferences, rather than simply restoring the image to its original form. Our focus is on transforming the scene captured in the photo to match subjective visual preferences, without relying on extensive labeled datasets.

2 Problem Formulation

We formulate the photo enhancement task as a Markov Decision Process (MDP) $\mathcal{M} = (\mathcal{S}, \mathcal{A}, \mathcal{T}, \mathcal{R}, \gamma)$, where an RL agent learns to predict a set of image processing parameters that improve the aesthetic quality of a given photograph. The state $s \in \mathcal{S}$ is represented as a concatenated feature vector inspired by [Shin et al., 2022], consisting of three components: an intensity feature f^i computed as a multi-scale histogram of the grayscale image, a gradient feature f^g computed as a multi-scale histogram of the Sobel edge map, and a semantic feature f^s extracted from a frozen pretrained ResNet-50 encoder. The final state vector is defined as:

$$s = [f^i, f^g, f^s] \quad (1)$$

The intensity and gradient histograms capture low-level photometric and structural properties of the image, while ResNet-50 provides high-level semantic context. All components are frozen during RL training to maintain a stable state representation, following the finding in [Shin et al., 2022] that learnable feature extractors lead to unstable training under sparse RL supervision.

At each episode, the agent observes the state and outputs a continuous four-dimensional action vector:

$$a = [\alpha, \beta, \delta_s, \gamma] \quad (2)$$

where $\alpha \in [0.5, 2.0]$ controls contrast scaling, $\beta \in [-50, 50]$ controls linear brightness shift, $\delta_s \in [-50, 50]$ controls saturation adjustment in HSV space, and $\gamma \in [0.5, 2.0]$ controls non-linear gamma correction which preferentially brightens darker regions. These parameters are applied to the input image using OpenCV to produce an enhanced output image s' . The reward is defined as the improvement in aesthetic quality measured by TOPIQ [Chen et al., 2024] (A Top-Down Approach from Semantics to Distortions for Image Quality Assessment), a no-reference image quality metric requiring no ground-truth reference:

$$r = \text{TOPIQ}(s') - \text{TOPIQ}(s) \quad (3)$$

The agent is trained using Proximal Policy Optimization (PPO) [Schulman et al., 2017] with a clipped surrogate objective to ensure stable policy updates. The optimization problem is defined as:

$$\max_{\theta} \mathbb{E}_{s \sim \mathcal{D}} [r(s, \pi_{\theta}(s))] \quad (4)$$

$$\text{subject to } \alpha \in [0.5, 2.0], \beta \in [-50, 50], \delta_s \in [-50, 50], \gamma \in [0.5, 2.0] \quad (5)$$

which is optimized via the clipped surrogate objective:

$$\mathcal{L}(\theta) = \mathbb{E}_t \left[\min \left(\rho_t(\theta) \hat{A}_t, \text{clip}(\rho_t(\theta), 1 - \epsilon, 1 + \epsilon) \hat{A}_t \right) \right] \quad (6)$$

where $\rho_t(\theta) = \pi_{\theta}(a_t|s_t)/\pi_{\theta_{\text{old}}}(a_t|s_t)$ is the probability ratio between the current and old policy, $\hat{A}_t$ is the estimated advantage, and ϵ is the clipping hyperparameter that prevents excessively large policy updates. For the dataset, we adopt an online training strategy that requires only a small initial collection of photographs $\mathcal{D} = \{I_1, \dots, I_N\}$. Rather than pre-generating a large set of augmented images, the agent explores enhancement trajectories by sampling actions and applying them to the input images via OpenCV on-the-fly during training, allowing the agent to encounter a diverse range of enhanced versions without any offline data preparation. We assume that (1) the TOPIQ score reliably approximates human aesthetic preference, and (2) the four chosen parameters are expressive enough to cover meaningful enhancement directions.

3 Empirical Evaluation

3.1 Benchmark Tasks & Datasets

We chose to use the MIT-Adobe FiveK Dataset [Bychkovsky et al., 2011], a collection of approximately 5000 images captured by professional photographers. This dataset provides high-quality images that are ideal for comparing the performance of image enhancement models.

3.2 Performance Metrics

To comprehensively and objectively compare the performance of our proposed method against other image enhancement and aesthetic improvement approaches under various conditions, we employ the following performance metrics:

3.2.1 Quantitative Comparison

Full Reference Metrics

These metrics are used when Ground Truth (GT) images are available to accurately measure the difference between the enhanced image and the original high-quality image. In our case, we adjust the images from the dataset (e.g., darkening, adding noise) and then use our model to generate the enhanced results. The original dataset images serve as the GT for comparison.

- **PSNR (Peak Signal-to-Noise Ratio):** Measures the pixel-level error between the enhanced image and the Ground Truth. A higher PSNR value indicates better fidelity of the image.
- **SSIM (Structural Similarity Index):** Measures the similarity between two images in terms of luminance, contrast, and structure. A higher SSIM value indicates greater similarity.

No Reference Metrics

Since Full Reference Metrics cannot fully represent human visual perception, we also use no-reference metrics to assess image quality and aesthetics. These metrics allow us to directly score the image and compare it with other baseline methods.

- **NIQE (Naturalness Image Quality Evaluator):** A pre-trained model based on the statistical features of natural images, which helps assess whether the structure and texture of the image match those of natural images. A lower NIQE score indicates better image quality.
- **TOPIQ (Topological Image Quality):** Unlike PSNR and SSIM, TOPIQ analyzes the image's structure based on topology and feature extraction, allowing it to evaluate the visual quality without requiring reference images.

3.2.2 User Study

Since objective metrics cannot fully capture subjective visual perception, we will use MOS (Mean Opinion Score) to have participants rate the enhanced images, assessing their aesthetic appeal, naturalness, detail preservation, and visual attractiveness.

3.3 Baseline Methods

We use the following papers, which are similar to our research objectives, as baselines for comparison:

- **DRL-ISP [Shin et al., 2022]:** This method uses Deep Reinforcement Learning for multi-objective Camera ISP enhancement.
- **StarEnhancer [Song et al., 2021]:** This method embeds stylistic features and utilizes a dual-AdaIN-based curve encoder to predict parameters for high-quality image enhancement.

4 Methodology

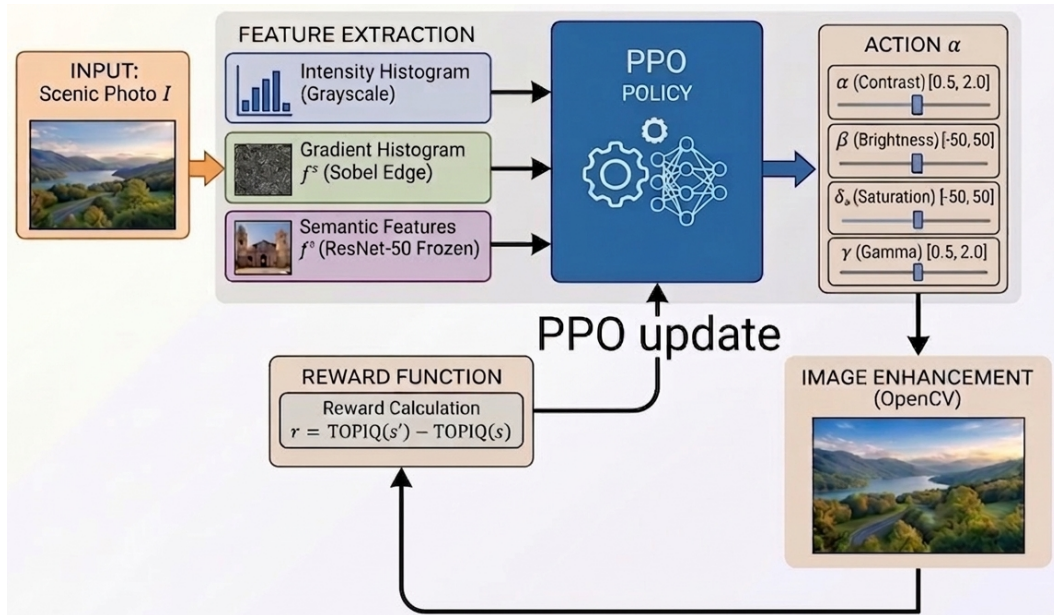


Figure 1: Overview of the reinforcement learning pipeline for scenic photo enhancement. An input image is processed to extract intensity, gradient, and semantic features, which form the state representation. The PPO policy predicts image adjustment parameters that are applied through an image enhancement module. The enhanced image is evaluated using the TOPIQ-based reward, and the policy is updated iteratively through PPO.

5 Experimental Result

6 Expected Contributions of Each Team Member

- **Chih-Hsuan, Chen:** Run baseline model and conduct various numerical evaluations for our model.
- **Shih-Chen, Tseng:** Implement PPO and work on reward updating based on [[Wu et al., 2024](#)].
- **Hsi-An, Chen:** Build CNN heuristic network, referencing [[Shin et al., 2022](#)] or [[Wu et al., 2024](#)].

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